In-Course Assessment Examination - 03

IT4123(T) Agent Computing

Department of Physical Science
University of Vavuniya

Time Allowed: **40 Minutes** Date: 19/12/2024

1.

a. Discuss the principles of coordination and cooperation in Multi-Agent Systems (MAS) and their role in solving distributed problems. [15%]

b: Explain with an example how can reinforcement learning techniques be applied to optimize agent interactions in competitive or cooperative MAS environments? [15%]

2. Discuss the importance of prompt engineering in optimizing performance in large language models. [15%]

3. Choose one of the following architectures: Transformer, RNN, or LSTM. Using your chosen model, explain how it can be applied to solve a text classification task (e.g., sentiment analysis). Address the following points:
   a) Describe the input representation and preprocessing steps required for the model. [10%]
   b) Outline the role of the key components in the model's architecture. [15%]
   c) Explain how the model learns to classify text into categories, by stating its advantages and limitations compared to the other two architectures. [15%]
   d) Suggest possible improvements or enhancements to the basic architecture to improve classification performance. [15%]

Dept. of Physical Science, CSH4162, IT4142, Compiler Design, ICA02, Dec 2024, 40 minutes

1. Describe three different definitions for a given grammar which is not of type LL(1). [15%]
2. Describe the model of a predictive parser with the aid of a diagram. [15%]
3. Consider a grammar with the following productions where S is the start symbol and \$ is the right end marker of the input:

   $S \rightarrow A\ B\ \$$
   $A \rightarrow x\ A$
   $A \rightarrow B$
   $B \rightarrow y\ z\ B$
   $B \rightarrow z$

   (a) Derive the sentence *xyzzz*\$ and show the parse tree. [10%]

   (b) Determine the FIRST and FOLLOW sets for each of the non-terminals of the grammar. [15%]

   (c) Construct the predictive parse table for the grammar. [15%]

   (d) Is this an LL(1) grammar? Justify your answer. [10%]

   (e) Suppose if we add a rule $A \rightarrow \varepsilon$, is the grammar still LL(1)? Justify your answer by constructing the parse table. [10%]

4. Produce the moves made by the predictive parser to parse the sentence *xyzzz*\$. [10%]

In-Course Assessment Examination - 01

IT4123(T) Agent Computing

Department of Physical Science
University of Vavuniya

Time Allowed: **40 Minutes** Date: 07/11/2024

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1. Define intelligent agents and discuss their main characteristics. Explain how these characteristics distinguish agents from standard software programs.

   [15%]

2. Consider a vehicle navigation system that can be implemented using different types of agents: Simple Reflex, Goal-Based, and Utility-Based.

   a. Compare and contrast how each type of agent would handle rerouting when an unexpected road closure occurs. [25%]

   b. Which agent type would provide the most efficient and user-friendly navigation experience in dynamic environments? Justify your answer. [15%]

3. Imagine you have been tasked with designing a MAS architecture to support a remote patient monitoring system for chronic diseases. The system must be capable of:

- Real-time monitoring: Continuously collect data from patients' wearable devices.
- Data processing and analysis: Analyze patient data to detect critical health events or trends that may require intervention.
- Alert and notification: Send alerts to healthcare providers and family members if there are any signs of a potential health risk.
- Data privacy and security: Ensure secure data communication and protect patients' personal health information.

a) Design a MAS architecture for the healthcare monitoring system, specifying the roles and responsibilities of each type of agent. [20%]

b) Define two *interaction protocols* for three agents and *data flow* among agents while sharing the information. [25%]

Department of Physical Science, CSH4162/IT4142, Compiler Design, ICA1, 40 minutes, Nov 2024

1. Define the four levels of *Chomsky*'s grammar classification. [20%]
2. State what is meant by parsing problem in compilation process. [05%]
3. State what is the disadvantage of an ambiguous grammar. [05%]
4. Derive the sentence `y+++y++` using left most and right most derivations: [10%]

```
S → A
A → A+A | B++
B → y
```

5. Consider a grammar with the following productions:

```
S → nS | AaB
A → x | xA | nA
B → y | yB | nB
```

   (a) Derive the sentence `nxayny` in all possible ways. [10%]

   (b) Identify whether the above grammar is ambiguous by using parse trees. [10%]

   (c) Write an equivalent unambiguous grammar to show that the same string `nxayny` can be derived from it. [15%]

6. Consider the regular expression `(x|y|z)aa*(x|y|z)`.

   (a) Construct NFA for the above regular expression. [10%]

   (b) Write down the productions for the above grammar. [15%]

Department of Physical Science

EL 4192 – Software Defined Networking

In-course Assessment – 01

Duration: 40 minutes **Date: 02-12-2024**

1. Explain the role of the OpenFlow protocol in SDN. [15 Marks]
2. Briefly discuss the key components of an OpenFlow-enabled switch. rule Action. [20 Marks]
3. What are the advantages and disadvantages of reactive and proactive flow setup in OpenFlow? [20 Marks]
4. What are the limitations of the OpenFlow protocol? [20 Marks]
5. Briefly describe the following control messages that are transmitted between SDN controller and OpenFlow switch? [20 Marks]
   - e. Packet_In
   - f. Packet_Out
   - g. Flow_Mod
   - h. Flow_ Removed

delivery

Department of Physical Science

Faculty of Applied Science

University of Vavuniya

IT 4133 Bioinformatics and Computational Biology (Practical)

In-course Assessment-02

Time Allowed: 75 Minutes **January 2025**

Instruction

All the relevant files are given in `bioinfor_ICAE2` *stored in your shared drive.*

1. Motifs are short, recurring sequences in DNA that are biologically significant. These could be transcription factor binding sites or other regulatory sequences. A motif is a substring of fixed length (k). For a sequence of length L, there are L-k+1 possible motifs. For each sequence, slide a window of size k to extract all motifs.

   For example (k=5 , seq1 = ATGCGTAGCTAAG):

   Motifs of length 5 are: ATGCG, TGCGT, GCGTA, CGTAG, GTAGC, TAGCT, AGCTA, GCTAA

   (a) Identify all possible motifs of length 5 for the sequences given in "seqs_for_motif.txt". [25%]

   (b) Count the frequency of each motif across all sequences. [20%]

   (c) Plot a bar graph of the most frequent motifs. [10%]

2. The "cell_line_case_control.csv" (dataset) file includes expression levels of genes across cell lines, with case-control status for each cell line.

   (a) Read the dataset. [10%]

   (b) Calculate the average expression level for each gene in cases and controls. [15%]

   (c) Identify genes that are significantly up-regulated in cases for each cell line (*Use your knowledge on Fold Change*). [15%]

   You are requested to use proper function definition, parameter passing, and function calls, along with appropriate comments [5%]

Note 1: Call each method at the end and print the results for testing purposes.

In-Course Assessment Examinations -01
EL4162 Numerical Computing(Theory)
Information Technology - 2023

Department of Physical Science

University of Vavuniya

Time Allowed: 1 hour Dec - 2024

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1. Convert the following binary numbers to decimal numbers [20%]

   (a) 1001010.01001_2 (b) -1100111.00111_2

2. Convert the following decimal numbers to binary numbers [20%]

   (a) 24.109375_{10} (b) 53.59375_{10}

3. Convert each of the following decimal numbers to equivalent octal and hexadecimal numbers: [20%]

   (a) -25.875_{10} (b) 18.109375_{10}

4. Draw the single precision floating point representation bits patterns of the the following decimal numbers. [40%]

   (a) 54.625
   (b) 63.046875

27

Department of Physical Science

IT4153 – Advanced Computer Networks

In-course Assessment Examination – 1

Time: 10.30 am

Date: 05.12.2024

1. Discuss the differences between IP and TCP header formats.
2. Explain how dynamic Network Address Translation (NAT) works.
3. Analyse the circuit switching technique used in WAN.
4. Describe what is meant by sliding window protocol.

1)

2)

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IT4133 Bioinformatics and Computational Biology (Theory)

In-course Assessment-01

Time Allowed: 45 Minutes **November 2024**

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codon chart is in back

1. You are studying a gene involved in a rare disease. You have both DNA-Seq and RNA-Seq data from patient and control samples. The goal is to identify mutations at the DNA level, see if these mutations affect RNA transcription, and analyze any resulting changes in the protein sequence. Given Sequences:

   - DNA sequence (patient): ATGGCGTAGTGA
   - DNA sequence (control): ATGGCGCAGTGA

   (a) Compare the DNA sequences from the patient and control to identify the mutation. Specify the type of mutation (e.g. substitution) and its location. [15%]

   (b) Using the patient's DNA sequence, transcribe it into RNA. [10%]

   (c) Using the RNA sequence from step 2, translate it into a protein sequence by dividing the RNA into codons (groups of three bases). Use the standard genetic code to determine the amino acid sequence. [10%]

   (d) Determine if the mutation in the patient's DNA leads to an amino acid change in the protein. If so, specify the original and mutated amino acids. [15%]

2. You are analyzing RNA-Seq data from two groups of samples: treated and untreated cells. You want to identify genes that are differentially expressed between the two conditions to understand how treatment affects gene activity.

   105 60 165 3 55

   - Gene A expression (Treated samples): [50, 55, 60]
   - Gene A expression (Untreated samples): [25, 30, 35]

   65 25 90

   (a) Calculate the mean expression level of Gene A in treated and untreated samples. [10%]

   55 30

(b) Determine the fold-change in Gene A's expression between treated and untreated cells. [15%]

(c) What does this suggest about Gene A's response to the treatment? [15%]

3. You perform scRNA-Seq on blood samples to study immune cell populations. Using clustering, you identify three clusters based on gene expression profiles. Average gene expression levels of the clusters for respected genes are given below

- Cluster 1: Gene A: 10, Gene B: 100, Gene C: 5
- Cluster 2: Gene A: 50, Gene B: 10, Gene C: 20
- Cluster 3: Gene A: 5, Gene B: 5, Gene C: 100

Marker genes details: Marker genes act as identifiers for different cell types

- Gene A: Marker for T cells
- Gene B: Marker for B cells
- Gene C: Marker for Natural Killer (NK) cells

(a) Assign a cell type to each cluster based on the marker gene expression. [10%]

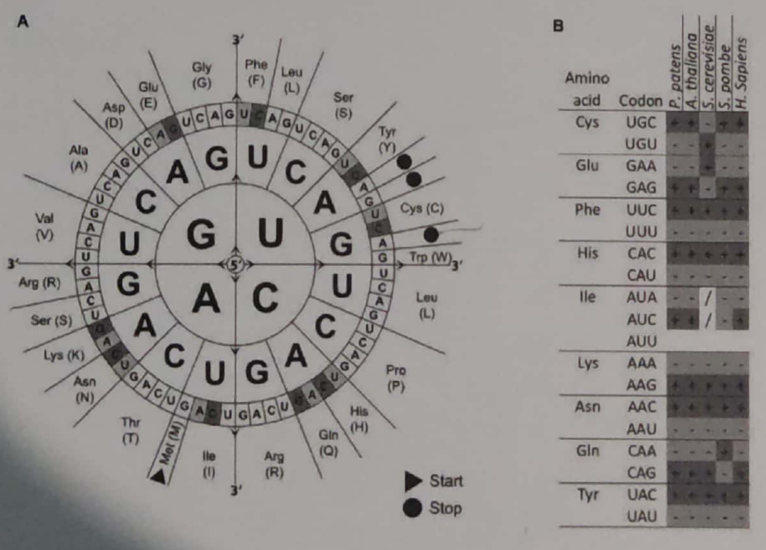

Figure 1: The Amino Acid/codons table

Note: A fold change > 1 indicates upregulation (increase) and downregulation if < 1.

$$\text{Fold Change} = \frac{\text{Expression in Condition A}}{\text{Expression in Condition B}}$$

In-Course Assessment Examination - 01

IT4123(P) Agent Computing

Department of Physical Science, Faculty of Applied Science
University of Vavuniya

Time Allowed: **60 Minutes**　　　　　　　　　　　　　　　　Date: 08/11/2024

key ["", "", "", ""]
values = data.split(",")
product = dict(zip(key, values))

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1. Develop an e-commerce chatbot agent for ***ShopEasy*** shopping site. The chatbot should initially collect product details and feedback from the command line, which will serve as its memory for later interactions with customers.

   (i). Prompt the user to input details of different products in the command line. For each product, collect:

   - **Product Name**
   - **Price**
   - **Stock Availability** (yes/no)
   - **User Ratings** (allow ratings between 1 and 5)

   This information should be stored in the chatbot's memory, which will be used to answer customer queries in real time.

   [30 marks]

   (ii). Allow the customer to ask only the following questions, which the chatbot will recognize and respond to:

   - "List of products?"
   - "Price of [product name]?"
   - "Is [product name] available?"
   - "What is the best-rated product?"
   - "Exit" to end the chat

   Display the chatbot's responses in a conversational format, as shown in the example below.

   [70 marks]

```
(base) C:\Users\ADMIN\Documents\project\ICA>py Server.py
Product data server started.

Enter product details. Type 'exit' to stop adding products.

Enter product name (or type 'exit' to stop): laptop
Enter price for laptop: $150
Is laptop available in stock? (yes/no): yes
Enter ratings for laptop (comma-separated between 1-5): 4
Product laptop added successfully!

Enter product name (or type 'exit' to stop): phone
Enter price for phone: $100
Is phone available in stock? (yes/no): yes
Enter ratings for phone (comma-separated between 1-5): 2,3,4,5,4,3
Product phone added successfully!

Enter product name (or type 'exit' to stop): tablet
Enter price for tablet: $80
Is tablet available in stock? (yes/no): yes
Enter ratings for tablet (comma-separated between 1-5): 4,2,3,4
Product tablet added successfully!

Enter product name (or type 'exit' to stop): exit

Waiting for client queries...
Client connected from ('127.0.0.1', 62721)
Customer: List of products
Chatbot: Available products: laptop, phone, tablet
Customer: Price of laptop
Chatbot: The price of laptop is $150.0
Customer: Is laptop available
Chatbot: Yes, laptop is currently in stock.
Customer: Best-rated product
Chatbot: Our best-rated product is laptop with an average rating of 4.0.
Customer: Exit
Chatbot: Thank you for using our service!

(base) C:\Users\ADMIN\Documents\project\ICA>
```

1. Server displaying chatbot responses.

```
(base) C:\Users\ADMIN\Documents\project\ICA>py Client.py
Welcome to ShopEasy! How can I help you today?
(Type 'exit' to end the chat)
Customer:  List of products
Chatbot: Available products: laptop, phone, tablet
Customer: Price of laptop
Chatbot: The price of laptop is $150.0
Customer: Is laptop available
Chatbot: Yes, laptop is currently in stock.
Customer: Best-rated product
Chatbot: Our best-rated product is laptop with an average rating of 4.0.
Customer: Exit
Chatbot: Thank you for using our service!
```

2. Client interacting with the chatbot

Department of Physical Science

Faculty of Applied Science

University of Vavuniya

IT4113 Computer Organization and Architecture (Theory)

In-course Assessment-01

Time Allowed: 30 Minutes November 2024

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1. What do you understand by computer architecture and organization? Explain in your own words.

2. What is **TSMC**, and what transistor technology and node size are planned for their latest transistors expected to be released in 2025?

3. Explain the steps involved in the execution of a program.

4. List the factors that influence the performance of a computer.

5. Given a computer system with four different instruction classes and their respective CPIs, answer the following questions. Assume the clock rate of the computer is 3 GHz.

   (a) If a program consists of 500 million instructions divided into the following instruction classes with the given distribution, calculate the total execution time:

   - Class A: 40% of instructions, CPI = 1.5
   - Class B: 25% of instructions, CPI = 2.0
   - Class C: 20% of instructions, CPI = 2.5
   - Class D: 15% of instructions, CPI = 3.0

   (b) Based on the distribution of instructions and CPI values for each instruction class, calculate the average CPI for the program.

   (c) Using the given clock rate and the total execution time calculated above, determine the total number of clock cycles required to complete the program.

Department of Physical Science
Faculty of Applied Science
University of Vavuniya

EL4142 – Graph Theory

In-Course Assessment Examination – I

Answer all Questions

Time: 40 minutes 26.11.2024

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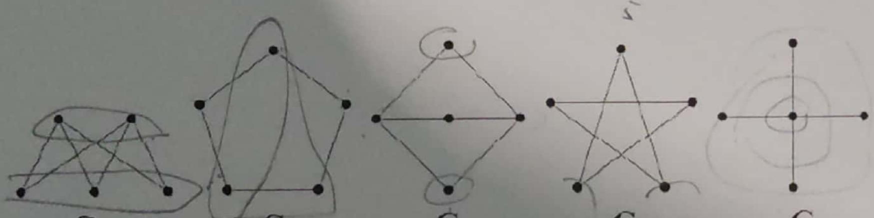
1. Define the following graphs with the aid of examples.
   i. Regular graph has the same no of degree
   ii. Complete graph each vertex connect with all the other vertices. [20%]

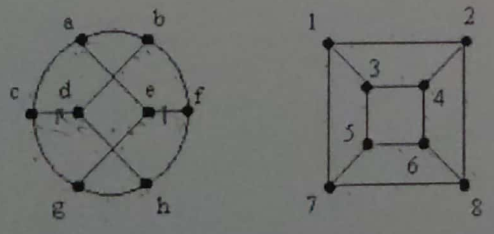
2. Let G = (V, E) be the graph with V(G) = {1,2,3,4,5,6} and E(G) = {(1,2), (1,4), (1,6), (2,3), (2,6), (4,5), (3,4), (5,6)}. Draw the graph G and write down the adjacency matrix A(G) and incidence matrix I(G). [15%]

3. Which of the following graphs is/are bipartite? Justify your answers.



G1 G2 G3 G4 G5 [20%]

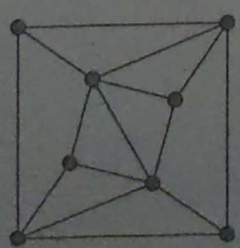
4. Determine whether the graphs G_1 and G_2 displayed below are isomorphic. justify your answer:

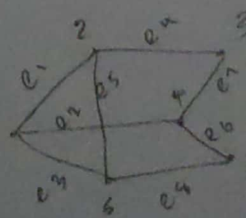


G1 G2 [15%]

5. Define walk, trivial, trail and path in a graph G. [15%]

6. Draw the complement of the following graph.





[15%]

Department of Physical Science

Faculty of Applied Science

University of Vavuniya

IT 4133 Bioinformatics and Computational Biology (Practical)

In-course Assessment-01

Time Allowed: 90 Minutes **December 2024**

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1. You are given the patient and control DNA sequences. These sequences also includes intron locations that must be removed (splicing) before translation. The relavant data and other details are given in `bioinfor_ICAE1_data.txt` stored in your shared drive. Write Python scripts to perform the following tasks on given DNA sequences:

   (a) Defines a method that can generate the antiparallel (complementary) DNA strand for both sequences. [15%]

   (b) Transcribe both sequences into RNA. [15%]

   (c) Splice both RNA sequences using given locations into matured mRNA. [20%]

   (d) Translate the RNA sequences into amino acid sequences using a provided codon-to-amino-acid dictionary. [25%]

   (e) Report the locations and differences (mutation) in amino acids between the two sequences. [15%]

   You are requested to use proper function definition, parameter passing, and function calls, along with appropriate comments [10%]

Note: Call each method at the end and print the results for testing purposes.

Department of Physical Science
Faculty of Applied Science
University of Vavuniya

EL4142 – Graph Theory

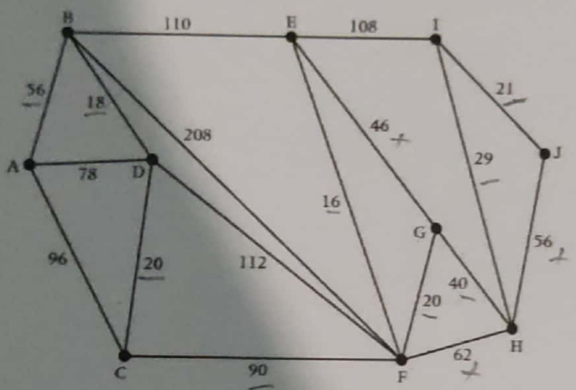
In-Course Assessment Examination – II

Answer all Questions

Time: 40 minutes

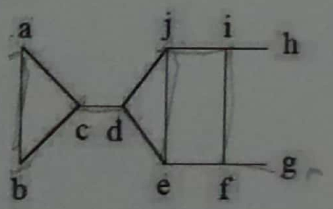
08.01.2025

1. Let G be a graph. Define the terms "Spanning tree of G" and "Cut edge of G". [20%]
2. The following graph shows a network of roads connecting ten towns. The number on each arc represents the distance, in miles, between two towns. Using Kruskal's algorithm, find the minimum spanning tree and minimum weight for the ten towns. [25%]

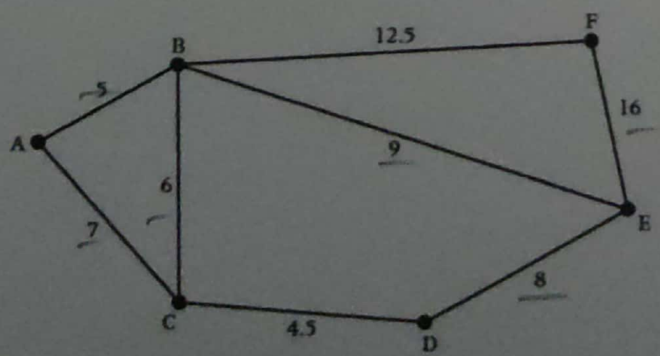


3
56
18
20
90
16
20
40
29
21
310

3. Using the Depth-first search and Bredth-first search algorithms, produce spanning trees of the following graph by choosing "a" as the root. Also find the hight of each spanning trees. [30%]



4. The following diagram shows the lengths of roads, in miles, that have to be gritted. The gritter is based at C and must travel along all the roads, at least once, before returning to C. Find an optimal Chinese postman route around the network, starting and finishing at C. State the length of the route. [25%]



12
6
18
9
27
16
43
8
51
4
55
12
67

68
12.5
80.5

CamScanner

IT 4153 (P): Practical for Advanced Computer Networks

ICAE 02

Department of Physical Science

Faculty of Applied Science, University of Vavuniya

Answer All Questions

Duration: 1 hour **07th of January 2025**

You are working as a trainee network engineer in a telecommunication company. Recently the company undertook a networking contract with a newly established IT company. The network estimation team has already gathered and documented the requirements for network design. The manager gave you the layout document for the simulation test using the packet tracer. The document represents the layouts of the buildings, network allocation and network devices.

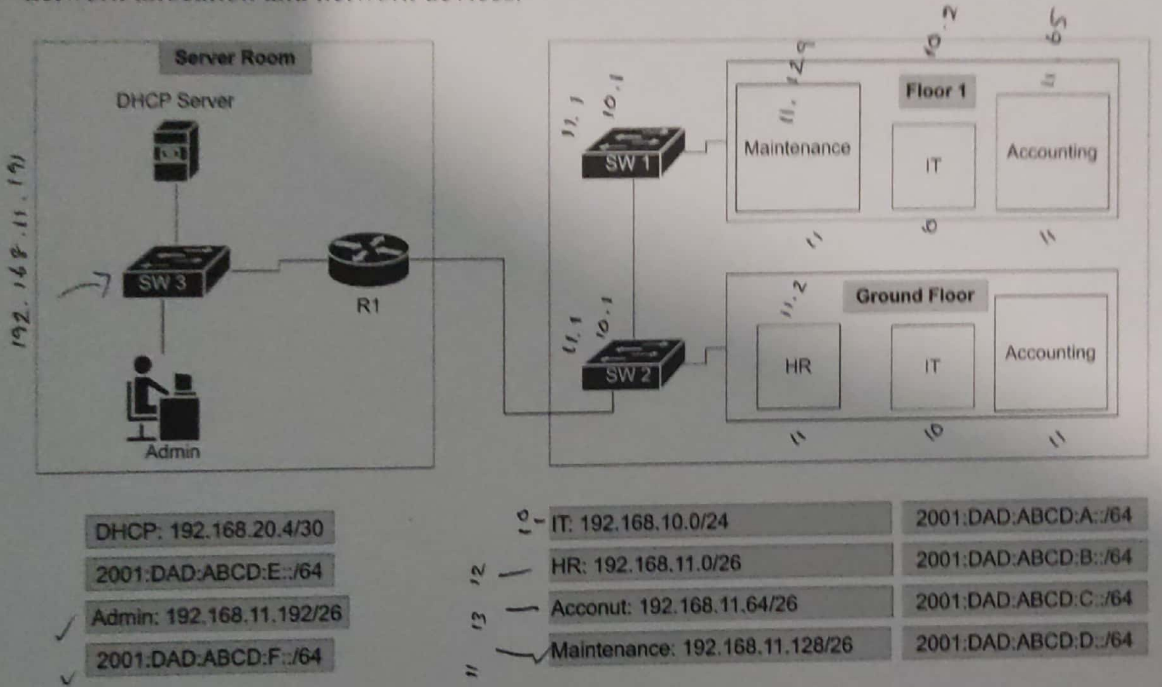


Figure 1 The Physical Network topology diagram of the company.

Figure 1 depicts the layout of the Server room and office space. The server room contains a router and a server which provides DHCP service to departments, in addition to that IP information is also mentioned in the figure.

1. Create a logical network topology for the buildings with proper labels and devices. You can use only a router and 3 switches in the building accordingly.
2. Each department belongs to a different network refer to Figure 1 for network information.
3. Use the dual-stack configuration method to assign IPv6 and IPv4 addresses, notably the routers should be configured with the first assignable address.
4. You must have at least two computers for each department, and it must get an IPv4 address from the external DHCP server and an IPv6 from the respective router as Stateful DHCPv6 Server. Configure 8.8.8.8 and 2001:DAD:CAFE:A::1 addresses for DNS in DHCP servers
5. The server room administrator should secure remote management on network devices using "class" or "cisco" as passwords and "admin" as the username.

In-Course Assessment Examinations -03

EL4162 Numerical Computing(Theory)

Information Technology - 2023

Department of Physical Science

University of Vavuniya

Time Allowed: 1 hour **Jan - 2025**

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1. Find the l_1, l_3 and l_∞ values for the following vectors. [45%]

(a) $x = \begin{bmatrix} 1 \\ -3 \\ 2 \end{bmatrix}$ (b) $y = \begin{bmatrix} 1 \\ -4 \\ 5 \end{bmatrix}$ (c) $z = \begin{bmatrix} 0.2 \\ -0.5 \\ 0.5 \\ -0.8 \end{bmatrix}$

2. Find the l_1 and l_∞ values for the following matrices. [30%]

(a) $A = \begin{bmatrix} 1 & -2 & 3 \\ -4 & 5 & -6 \\ 7 & -8 & 9 \end{bmatrix}$ (b) $B = \begin{bmatrix} 1 & -1 & 0 \\ 2 & 3 & -1 \\ 5 & 0 & -2 \end{bmatrix}$ (c) $C = \begin{bmatrix} 1 & 3 & 2 & 5 \\ 2 & 4 & 1 & 7 \\ 1 & 2 & 1 & 9 \\ 6 & 8 & 3 & 5 \end{bmatrix}$

3. Let $\begin{pmatrix} -2 & 1 \\ 2 & 0 \end{pmatrix}$ [25%]

(a) Find $\frac{||Ax||_1}{||x||_1}$ for the following three vectors. $x_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$, $x_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$, $x_3 = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$

(b) Find a vector x such that $\frac{||Ax||_1}{||x||_1} = ||Ax||_1$

Department of Physical Science
Faculty of Applied Science
University of Vavuniya

EL4142 – Graph Theory

In-Course Assessment Examination – III

Answer all Questions

Time: 40 minutes 15.01.2025

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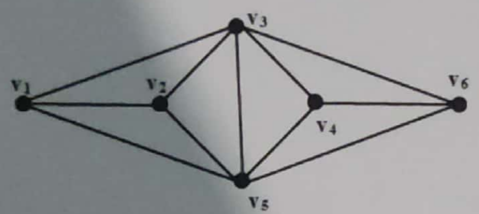
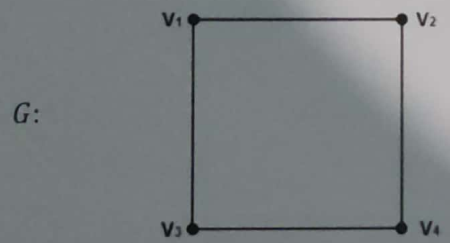
1. Define a Planar graph and a Dual graph of a planar graph. [20%]
2. If a planar graph G with 7 vertices divides the plane into 8 faces, how many edges must G have? [15%]
3. Draw the dual graph for the following graph. [10%]



4. Draw the line graph L(G), $L^2(G)$ and total graph for the following graph G. [30%]



5. A tropical fish hobbyist has six different types of fish: Alphas, Betas, Certas, Deltas, Epsas and Fetas, which are denoted by A, B, C, D, E, and F respectively. Because of predator-prey relationships, water conditions and size some fish cannot be kept in the same tank. The following table shows which fish cannot be together. Fish Type Cannot be with:

Fish Type	Cannot be with
A	B C
B	A C E
C	A B D E
D	C F
E	B C F
F	D E

   i. Represent the above problem into a graph. [10%]
   ii. What is the smallest number of tanks needed to keep all of the fish? [15%]